Nivedita Taneja

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Work Experience

Amazon Inc. May 2025 – Aug 2025
Software Development Engineer Intern – Devices (ACRS)
Tech Stack: Java, Python, Spring Boot, Large Language Models (LLMs), CLI Development, MCP, AWS
- Developed an **AI-powered code cleanup engine** using **Python** and **Java + React** for **micro frontends** to automate deprecation of feature flags (and weblabs) configured in highly inconsistent ways across Amazon services, addressing edge cases like nested conditionals, overrides, and partial rollouts, delivering 10x faster deprecations in practice
- Validated at org scale, driving a reduction of **90% in time** (4 hours to 20 min) and **99% in cost** ($600 to $6) per cleanup
- Built around an **LLM-based** analysis and transformation core, solving challenges like large-file context window limits with segmentation strategies such as splitting code into logical units, sliding-window chunking, context preservation, hierarchical summarization of segments, custom prompts per code pattern, and semantic preservation, achieving **91% first-pass accuracy**
- Implemented a custom **fault-tolerant** validation and auto-recovery pipeline with syntax/compile checks, automated unit test updates with coverage guarantees, and intelligent error-specific retries to ensure correctness and resilience of the transformed code, reducing manual corrections to less than 10% of cases

Broadcom Inc. Apr 2023 – Jul 2024
Member of Technical Staff – End User Computing (WorkspaceONE)
Tech Stack: Angular, Typescript, Node.js, Java, Ruby on Rails, Microservices, SQL, Cloud Computing
- Reengineered legacy admin experience UI with **full-stack** ownership **(Angular/Node.js)**, incorporating global search/filter and device interaction flows, optimized async rendering and state management for a **27% faster, more interactive** experience
- Developed fault-tolerant **microservices** in **Node.js/Java** with **SQL** backends, scaling to millions of device management requests daily and improving throughput and availability by 33%
- Architected session management and caching layers with resiliency and observability hooks, integrating **API Gateway** metrics and SLIs to provide reliable infra-level visibility across services
- Optimized multi-tenant RBAC and authentication flows in **Ruby on Rails web services**, reducing query latency by 16% and aligning security enforcement with enterprise policy requirements

VMware Inc. Jan 2022 – Apr 2023
Member of Technical Staff – VMware Blockchain (OCTO)
Tech Stack: Go, React, Python, Kubernetes, AWS, PostgreSQL, Multithreading, Distributed Systems, gRPC
- Developed and deployed the Transaction Flow Service using **Ruby on Rails** with **distributed PostgreSQL** clusters, reducing transaction execution time by 28% in blockchain SaaS workloads
- Instituted a **multithreaded** testing framework in **Python and Go** with intelligent fault-injection, load simulation, and health monitoring, lowering triage time by 15 hours/week and cutting flaky failures by 42%
- Set up a containerized infrastructure with **Kubernetes** and **AWS**, using **Infrastructure as Code (Helm/Terraform-style patterns)** for scalable deployment and CI pipelines and multi-region resiliency, cutting deployment times by 55%
- Created ChainViz blockchain state visualizer, a **Grafana-style** real-time observability tool visualizing transaction flows and system metrics, backed by the **GraphQL** endpoint; enhanced debugging efficiency by 38%, supporting internal SLI/SLO targets, adopted **persisted queries** and cache hints to keep dashboards responsive under load

Optum Global Solutions, United Health Group June 2021 – August 2021
Software Engineering Intern
Tech Stack: .NET, C#, MVC, Nunit, Specflow, Selenium, Cloud QA, Regression Testing, Integration Testing, BDD
- Architected an monitoring-aware **acceptance test-driven development (ATDD) framework** employing **SpecFlow** and **Selenium**, enabling automated regression detection and proactive log-based alerts, increasing test reliability across environments
- Developed an automatic data purging system for automation-related data, lowering infra storage usage by 40% and ensuring test environment hygiene in nightly regression runs.
- Designed **.NET Core and C#** based tooling for indexing, linking, and viewing QA artifacts, boosting document handling and workflow management throughput by 35% for automation analysts

CodeReady.Inc - Canadian EdTech startup founded by IIT Alumni September 2020 – June 2021
Software Developer
Tech Stack: React, TypeScript, Node.js, Express, Redis, AWS, Docker, Cypress/Playwright, HTML/CSS
- Implemented a **registration & scheduling portal** in **React + Node.js** with atomic slot reservations, duplicate detection, and server-side validation; integrated **SES** reminders with SPF/DKIM/DMARC alignment and **.ics** calendar attachments
- Developed **REST APIs** for courses/sessions/enrollments (**Express/PostgreSQL**) with schema validation, idempotency keys, and **Redis** caching; JSON logs & backpressure-aware workers for bursts, reducing **P99 latency** and 5xx error rates by **42%**
- Optimized front-end via **route-level code-splitting**, image lazy-loading, and **HTTP** caching; enforced **TypeScript ESLint** rules, **Lighthouse** budgets in **CI**. Improved LCP 38% and TBT 44% on low-end Chromebooks and older iPads used in classrooms
- Containerized services with **Docker** and implemented **CI/CD** via **GitHub Actions**; introduced metrics, tracing, and structured logs with alerting & runbooks, cutting **MTTR by 39%** and release lead time by 49%

Technical Skills

Programming Languages : Python, C++, Java, Javascript/TypeScript, Go, Ruby, C#, Bash
Web Development : React, Angular, Typescript, ExpressJS, Node.js, Spring Boot, Microservices, Webservices, .NET, GraphQL
Databases : PostgreSQL, MySQL Database, NoSQL (MongoDB, Redis, Cassandra), HDFS (Hadoop Distributed File System)
Cloud Infrastructure : Amazon Web Services (AWS), Docker, Kubernetes, Google Cloud Platform (GCP), Microsoft Azure
Monitoring & Reliability : Grafana, Prometheus, Fault tolerance, SLI/SLO, Resilient systems, High availability, Observability
Testing & CI/CD : JUnit, NUnit, SpecFlow, Selenium, Cypress, Playwright, Jenkins, Terraform

Education

San Jose State University San Jose, United States
*Masters of Science in Software Engineering — GPA - **4.0*** *August 2024 – May 2026*
Thapar Institute of Engineering and Technology Patiala, India
*Bachelors of Technology in Computer Science and Engineering — GPA - **9.14*** *July 2018 – May 2022*